Model-free Prediction

Recall

- **Goal**: learn $v_\pi(s)$ from episodes of experience under policy π,

- Data: $S_1, A_1, R_2, S_2, A_2, \cdots, S_T \sim \pi.$

- Return: $G_t = R_{t+1} + \gamma R_{t+1} + \cdots + \gamma^{T-1} R_T$

- Value function

$$\begin{cases} v_\pi(s) = E_\pi(G_t \mid S_t = s) \\ q_\pi(s,a) = E_\pi(G_t \mid S_t = s,\ A_t = a) \end{cases}$$

Example: Blackjack

(21点)

(youtube)

⇓

Homework

- Known MDP

  Planning by Dynamic Programming

  ① Evaluate $v_\pi(s) / q_\pi(s,a)$ by Bellman

  ② $\pi(a|s) = \arg\max_a q_\pi(s,a)$

- Unknown MDP.

  ① Evaluate $v_\pi(s) / q_\pi(s,a)$ by

  { Monte Carlo
  { Temporal Difference (TD)

  ② $\pi(a|s) = \arg\max_a q_\pi(s,a)$

Monte-Carlo Prediction

First-Visit MC prediction: $V \approx v_\pi$.

- For first visit $S_t = s$ in each episode,

$$S_0^{(1)}, A_0^{(1)}, R_1^{(1)}, S_1^{(1)}, R_2^{(1)}, \cdots, S_{T-1}^{(1)}, A_{T-1}^{(1)}, R_{T_1}^{(1)}$$
$$\text{s:} \quad G^{(1)}(s) = R_2^{(1)} + \gamma R_3^{(1)} + \cdots +$$

$$\vdots$$

$$S_0^{(m)}, A_0^{(m)}, R_1^{(m)}, S_1^{(m)}, R_2^{(m)}, \cdots, S_{T-1}^{(m)}, A_{T-1}^{(m)}, R_{T_m}^{(m)}$$
$$\text{s:} \quad G^{(m)}(s) = R_{T_m}^{(m)}$$

$$\vdots$$

$$S_0^{(M)}, A_0^{(M)}, R_1^{(M)}, S_1^{(M)}, R_2^{(M)}, \cdots, S_{T-1}^{(M)}, A_{T-1}^{(M)}, R_{T_M}^{(M)}.$$

$$\Rightarrow \quad V(s) = \frac{1}{M} \sum_{m=1}^{M} G^{(m)}(s)$$

$$\text{or} \quad V(S_t) = \frac{1}{M} \sum_{m=1}^{M} G^{(m)}(S_t)$$

Alternatively.

- $S_0, A_0, R_1, S_1, R_2, \cdots, S_t, A_t, R_{t+1}, \cdots, S_{T-1}, A_{T-1}, R_T$

$V(S_{T-1}) = R_T$

$V(S_t) = R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{T-t-1} R_T$

$V(S_1) = R_2 + \gamma R_3 + \cdots + \gamma^{T-2} R_T$

$V(S_0) = R_1 + \gamma R_2 + \cdots + \gamma^{T-1} R_T$

- $\hat{v}_\pi^{MC}(S_t) = \text{average}[V(S_t)]$

First-visit MC prediction, for estimating $V \approx v_\pi$

Input: a policy π to be evaluated

Initialize:

$V(s) \in \mathbb{R}$, arbitrarily, for all $s \in \mathcal{S}$

$Returns(s) \leftarrow$ an empty list, for all $s \in \mathcal{S}$

Loop forever (for each episode):

Generate an episode following π: $S_0, A_0, R_1, S_1, A_1, R_2, \ldots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

Loop for each step of episode, $t = T-1, T-2, \ldots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

Unless S_t appears in $S_0, S_1, \ldots, S_{t-1}$: → i.e. first visit

Append G to $Returns(S_t)$

$V(S_t) \leftarrow \text{average}(Returns(S_t))$

Monte-Carlo Policy Evaluation

- To evaluate state s
- The first time-step t that state s is visited in an episode
- Increment counter $N(s) \leftarrow N(s) + 1$
- Increment total return: $S(s) \leftarrow S(s) + G_t$
- Mean estimate: $V(s) = S(s)/N(s)$
- SLLN: $V(s) \xrightarrow{a.s} v_\pi(s)$ as $N(s) \to \infty$.

Every-Visit MC prediction: $V \approx v_\pi$.

- For every $S_t = s$ in each episode.

s: $G^{(1)}(s) = R_2^{(1)} + \gamma R_3^{(1)} + \cdots +$

$S_0^{(1)}, A_0^{(1)}, R_1^{(1)}, S_1^{(1)}, R_2^{(1)}, \cdots, S_{T-1}^{(1)}, A_{T-1}^{(1)}, R_{T_1}^{(1)}$

s: $G^{(2)}(s) = R_{T_1}$

$\vdots$

$S_0^{(m)}, A_0^{(m)}, R_1^{(m)}, S_1^{(m)}, R_2^{(m)}, \cdots, S_{T-1}^{(m)}, A_{T-1}^{(m)}, R_{T_m}^{(m)}$

s: $G^{(m)}(s) = R_{T_m}^{(m)}$

$\vdots$

$S_0^{(M)}, A_0^{(M)}, R_1^{(M)}, S_1^{(M)}, R_2^{(M)}, \cdots, S_{T-1}^{(M)}, A_{T-1}^{(M)}, R_{T_M}^{(M)}$.

$$\Rightarrow \quad V(s) = \frac{1}{N} \sum_{n=1}^{N} G^{(n)}(s)$$

Monte-Carlo Policy Evaluation

- To evaluate state s
- The every time-step t that state s is visited in an episode
  - Increment counter $N(s) \leftarrow N(s) + 1$
  - Increment total return: $S(s) \leftarrow S(s) + G_t$
  - Mean estimate: $V(s) = S(s)/N(s)$
  - SLLN: $V(s) \xrightarrow{a.s} v_\pi(s)$ as $N(s) \to \infty$.

Monte-Carlo Prediction

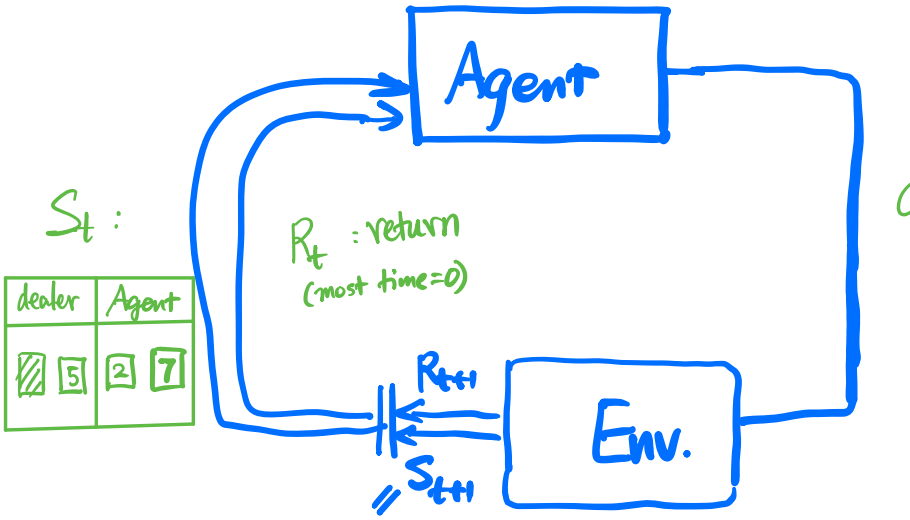
- Learns directly from episodes of experience.
- Model-free: No knowledge of MDP.
- Learns from complete episodes
  (not part of it, as in TD)
- Forward-view
- Sample means $\xrightarrow{\text{SLLN}}$ population means.
- Caveat:

  only apply to episodic MDPs.

Example: Blackjack

Episodic Game:

stop if $G_t = \begin{cases} \$1 & \text{win} \\ \$0 & \text{draw} \\ \$-1 & \text{lose} \end{cases}$



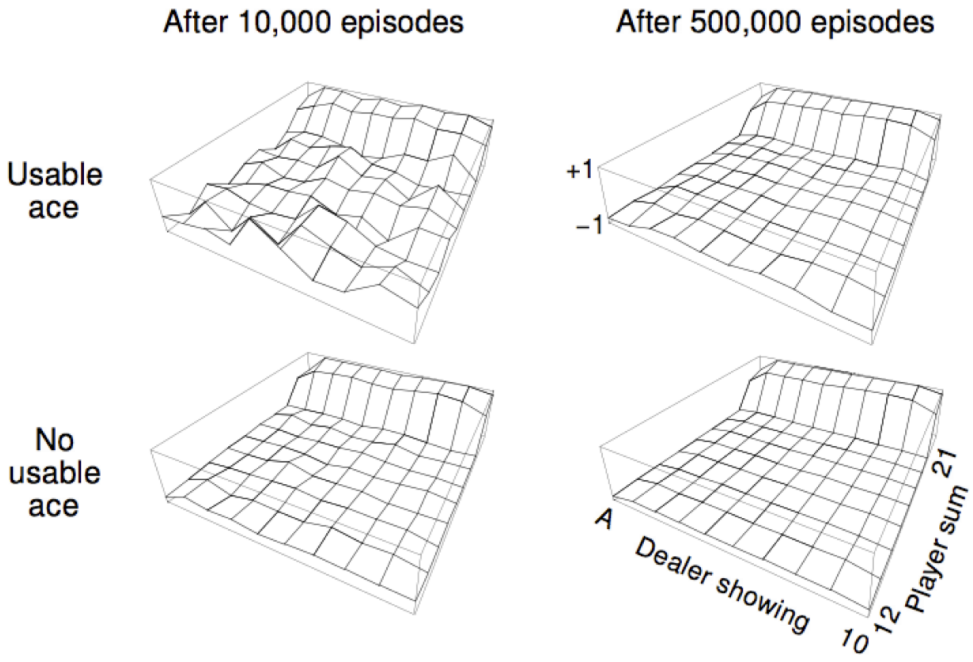
- Agent: Cards ~ {1, ..., 10, 11}
- Dealer: respond → hit if ≤ 16
  → stick if > 16 (no choice)

(S, A, R, P, γ)

- States (200 of them):
  - Current sum (12-21)
  - Dealer's showing card (ace-10)
  - Do I have a "useable" ace? (yes-no)
- Action stick: Stop receiving cards (and terminate)
- Action twist: Take another card (no replacement)
- Reward for stick:
  - $+1$ if sum of cards $>$ sum of dealer cards
  - 0 if sum of cards $=$ sum of dealer cards
  - -1 if sum of cards $<$ sum of dealer cards
- Reward for twist:
  - -1 if sum of cards > 21 (and terminate)
  - 0 otherwise
- Transitions: automatically twist if sum of cards < 12



Blackjack Value Function after Monte-Carlo Learning



Policy: stick if sum of cards ≥ 20, otherwise twist

↓↓

Bad policy!

How to improve it?

Incremental MC update

Sample	Mean
• x_1	$\mu_1 = x_1$
• $x_1, x_2,$	$\mu_2 = \frac{1}{2}(x_1 + x_2)$ $= \mu_1 + \frac{1}{2}(x_2 - \mu_1)$
• $x_1, x_2, x_3,$	$\mu_3 = \frac{1}{3}(x_1 + x_2 + x_3)$ $= \frac{1}{3}(2\mu_2 + x_3)$ $= \mu_2 + \frac{1}{3}(x_3 - \mu_2)$



- $X_1, \cdots X_k, \quad \mu_k = \frac{1}{k}(X_1 + \cdots + X_k)$

$$= \frac{1}{k}\left((k-1)\mu_{k-1} + X_k\right)$$

$$= \mu_{k-1} + \frac{1}{k}\left(X_k - \mu_{k-1}\right)$$

Incremental MC update

- Update $V(s)$ incrementally after episode

$S_0, A_0, R_1, S_1, R_2, \cdots, S_t, A_t, R_{t+1}, \cdots, S_{T-1}, A_{T-1}, R_T$

- For each state S_t with return G_t.

$$N(S_t) \leftarrow N(S_t) + 1$$

$$V(S_t) \leftarrow V(S_t) + \frac{1}{N(S_t)}(G_t - V(S_t))$$

- In practice.

$$V(S_t) \leftarrow V(S_t) + \alpha_t(G_t - V(S_t))$$

For example.

- $\alpha_t \equiv \alpha$

$$V(S_t) \leftarrow V(S_t) + \alpha\,(G_t - V(S_t))$$

$$y_{t+1} = y_t + \alpha(G_t - y_t)$$

$$= \alpha G_t + (1-\alpha)\, y_t$$

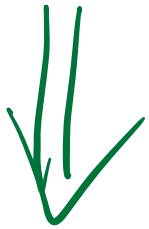
$$= \alpha G_t + (1-\alpha)\left[\alpha G_{t-1} + (1-\alpha)\, y_{t-1}\right]$$

$$= \alpha G_t + \alpha(1-\alpha) G_{t-1} + (1-\alpha)^2 y_{t-1}$$

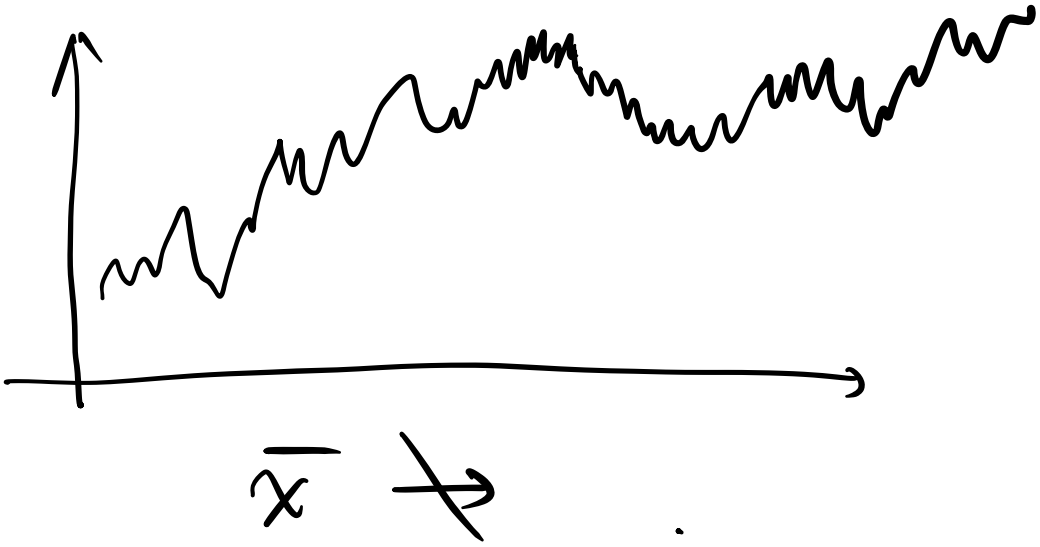
$$= \alpha G_t + \alpha(1-\alpha) G_{t-1} + \alpha(1-\alpha)^2 G_{t-2} + (1-\alpha)^3 y_{t-3}$$

$$= \alpha\left[G_t + (1-\alpha) G_{t-1} + (1-\alpha)^2 G_{t-2} + \cdots\right]$$

i.e Weighted averages with exponentially decaying weights



Good for non-stationary problems,



- $\alpha_t \searrow 0$, e.g.

Robbins-Monro algorithm:

① $\sum \alpha_t = \infty$

② $\sum \alpha_t^2 < \infty$

i.e GLIE.

Temporal Difference Prediction (TD)

- Episode:

$$S_0, A_0, R_1, S_1, A_1, R_2, \cdots,$$

$$\cdots, S_t, A_t, R_{t+1}, \cdots S_{T-1}, A_{T-1}, R_T.$$

- MC:

$$V(S_t) \leftarrow V(S_t) + \alpha \left(G_t - V(S_t)\right)$$

$$G_t = R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{T-t-1} R_T$$

$$= R_{t+1} + \gamma G_{t+1}$$

⇓

Need complete episode

- TD :

$$V(S_t) \leftarrow V(S_t) + \alpha (G_t - V(S_t))$$

$$G_t = R_{t+1} + \gamma V(S_{t+1})$$

$$\Downarrow$$

Only need

$$\{S_t, A_t, R_{t+1}, S_{t+1}\}$$

$$\Uparrow$$

experience.

Mini Summary on TD

- Goal:

  learn v_π online from experience under policy π.

- TD:

  Update $V(S_t)$ toward estimated return $R_{t+1} + \gamma V(S_{t+1})$.

$$V(S_t) \leftarrow V(S_t) + \alpha \underbrace{\left[\underbrace{R_{t+1} + \gamma V(S_{t+1})}_{\text{TD Target}} - V(S_t) \right]}_{\delta_t : \text{TD error}}$$

TD

- Learning directly from episodes of experience
- Model-free:

  No knowledge of MDP
- Learns from incomplete esipodes $(S_t, A_t, R_{t+1}, S_{t+1})$

  i.e. "bootstrapping"

  ⇓
- Update a guess from a guess

Advantages and Disadvantages of MC vs. TD

- TD can learn *before* knowing the final outcome
  - TD can learn online after every step
  - MC must wait until end of episode before return is known
- TD can learn *without* the final outcome
  - TD can learn from incomplete sequences
  - MC can only learn from complete sequences
  - TD works in continuing (non-terminating) environments
  - MC only works for episodic (terminating) environments

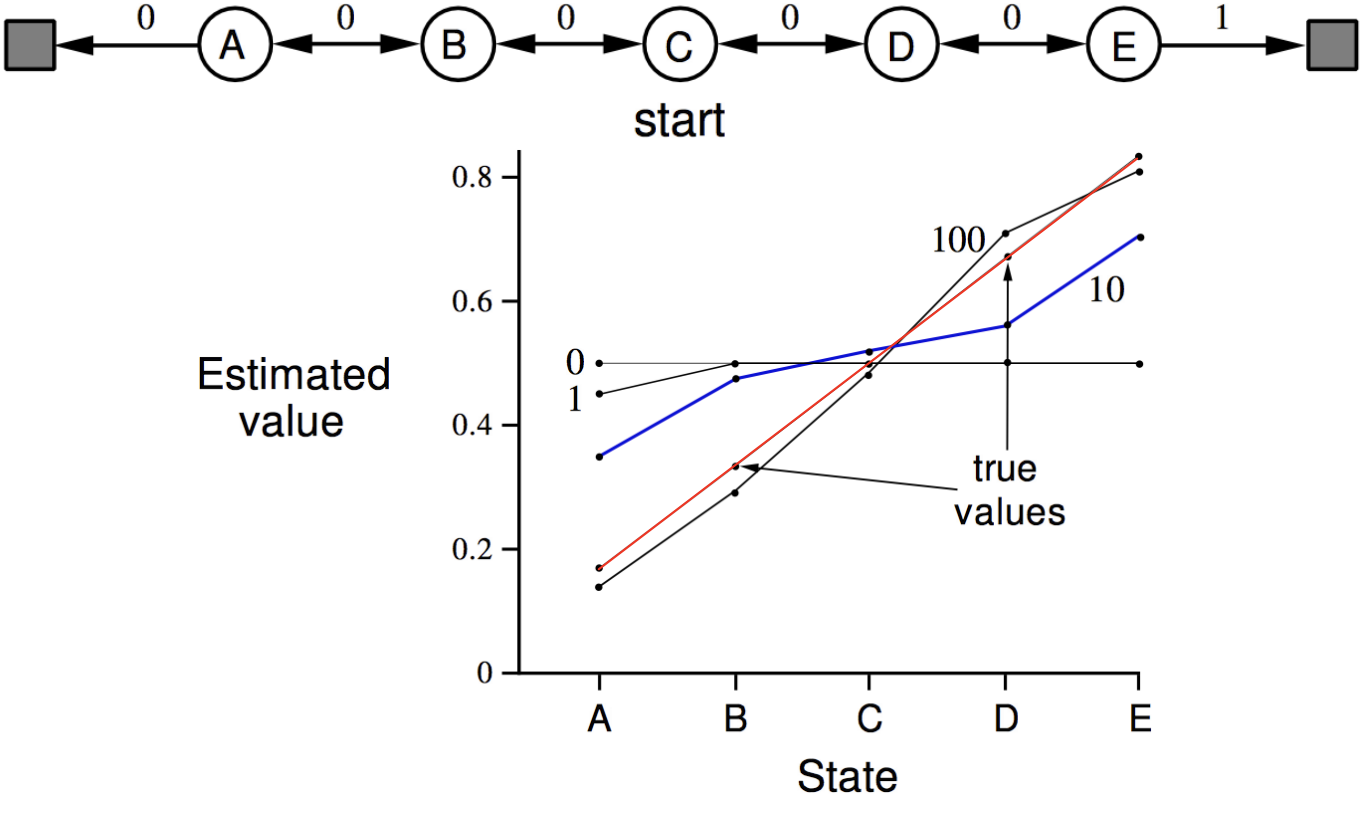
Bias/Variance Trade-Off

- Return $G_t = R_{t+1} + \gamma R_{t+2} + ... + \gamma^{T-1} R_T$ is *unbiased* estimate of $v_\pi(S_t)$
- True TD target $R_{t+1} + \gamma v_\pi(S_{t+1})$ is *unbiased* estimate of $v_\pi(S_t)$
- TD target $R_{t+1} + \gamma V(S_{t+1})$ is *biased* estimate of $v_\pi(S_t)$
- TD target is much lower variance than the return:
  - Return depends on *many* random actions, transitions, rewards
  - TD target depends on *one* random action, transition, reward

Advantages and Disadvantages of MC vs. TD (2)

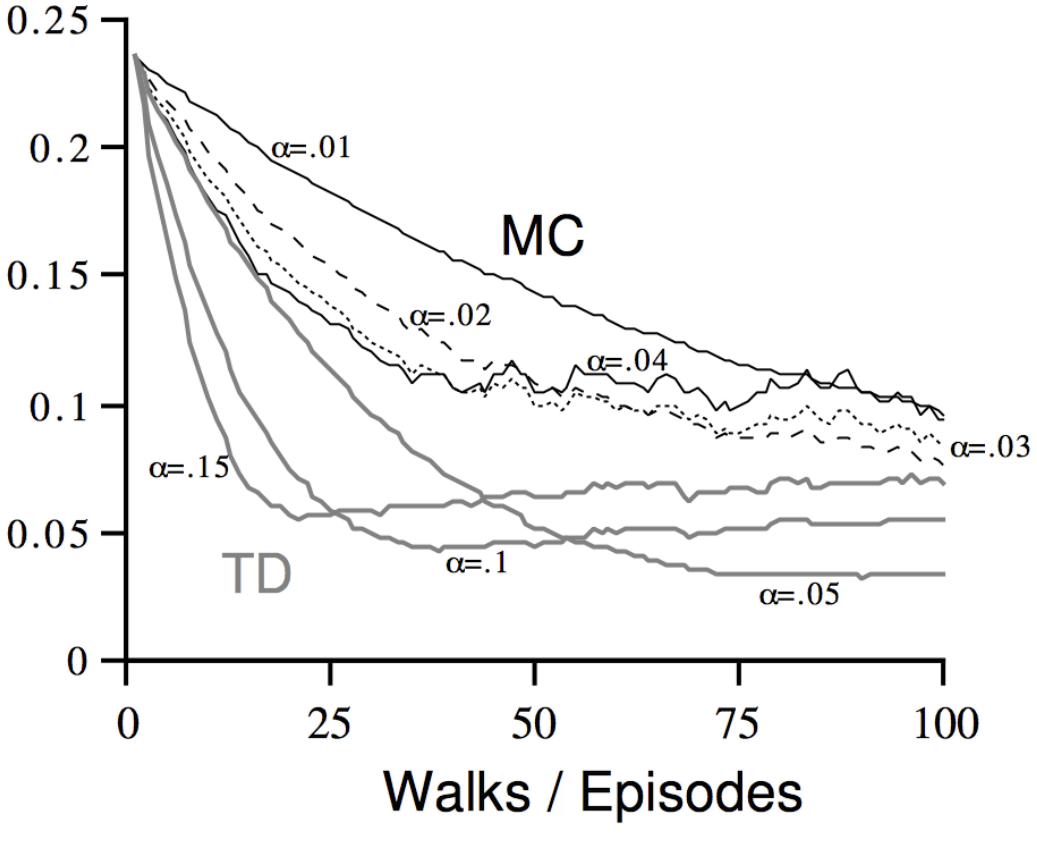
- MC has high variance, zero bias
  - Good convergence properties
  - (even with function approximation)
  - Not very sensitive to initial value
  - Very simple to understand and use
- TD has low variance, some bias
  - Usually more efficient than MC
  - TD(0) converges to $v_\pi(s)$
  - (but not always with function approximation)
  - More sensitive to initial value

Random Walk Example



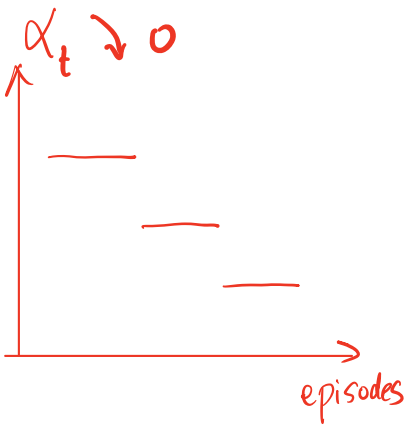
Random Walk: MC vs. TD

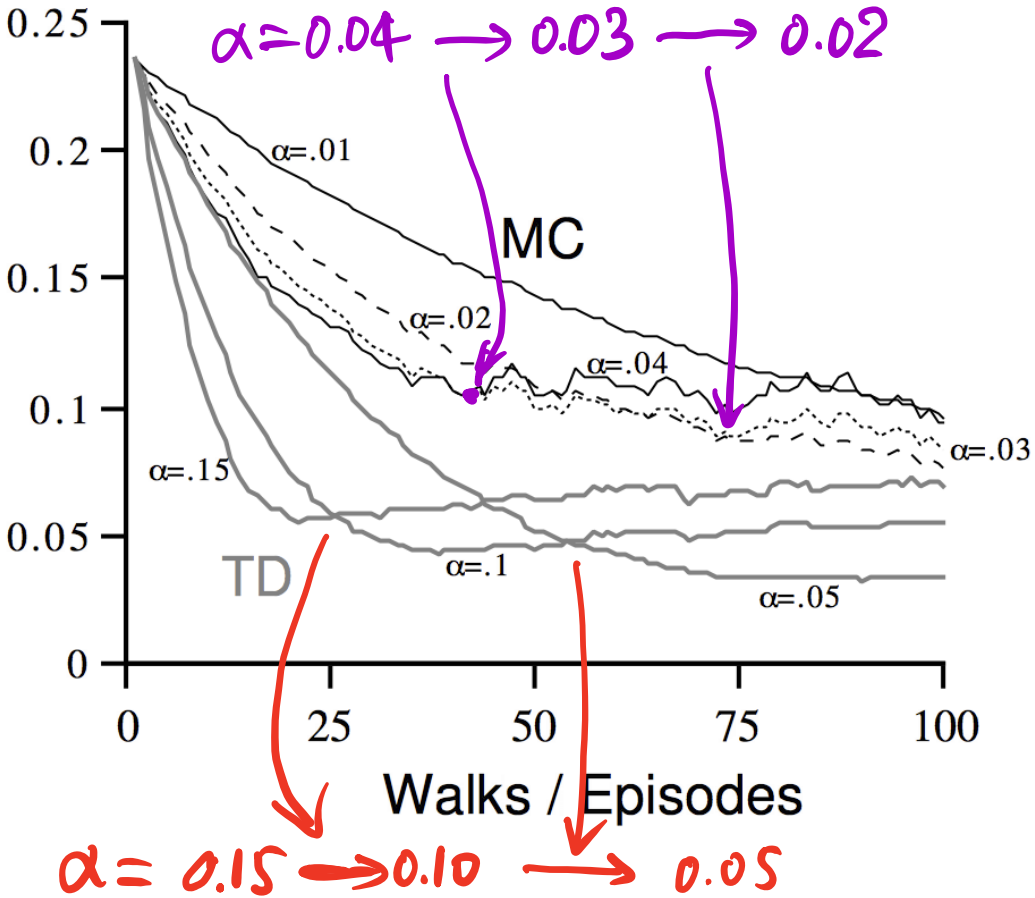
RMS error, averaged over states



Random Walk: MC vs. TD

RMS error, averaged over states





Batch MC and TD

- MC and TD converge: $V(s) \to v_\pi(s)$ as experience $\to \infty$
- But what about batch solution for finite experience?

$$s_1^1, a_1^1, r_2^1, ..., s_{T_1}^1$$
$$\vdots$$
$$s_1^K, a_1^K, r_2^K, ..., s_{T_K}^K$$

  - e.g. Repeatedly sample episode $k \in [1, K]$
  - Apply MC or TD(0) to episode k

AB Example

Two states A, B; no discounting; 8 episodes of experience

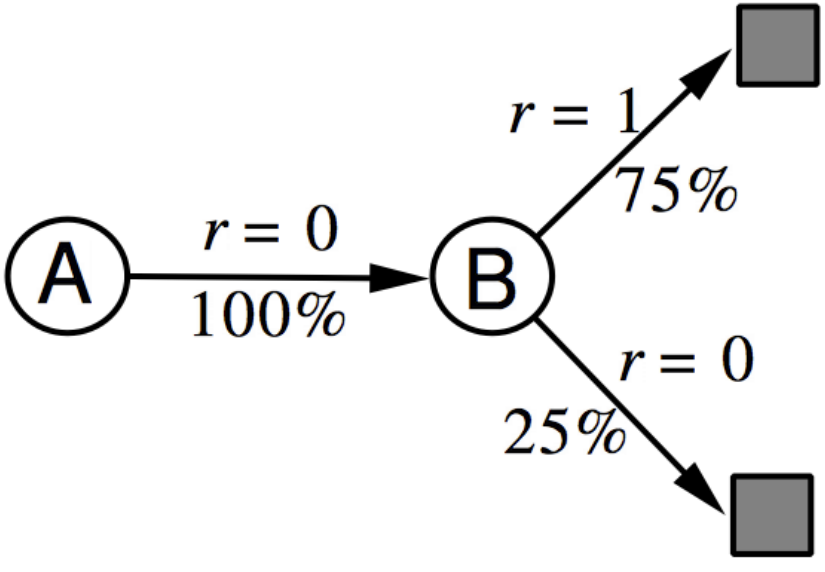
$$\mathrm{A}, 0, \mathrm{B}, 0$$
$$\mathrm{B}, 1$$
$$\mathrm{B}, 1$$
$$\mathrm{B}, 1$$
$$\mathrm{B}, 1$$
$$\mathrm{B}, 1$$
$$\mathrm{B}, 1$$
$$\mathrm{B}, 0$$

What is $V(A), V(B)$?

AB Example

Two states A, B; no discounting; 8 episodes of experience

$A, 0, B, 0$
$B, 1$
$B, 1$
$B, 1$
$B, 1$
$B, 1$
$B, 1$
$B, 0$



What is $V(A), V(B)$?

- Monte-Carlo.

$$V(A) = 0.$$

$$V(B) = 6/8 \quad \text{First/Every visit.}$$

- Alternative writing:

$$\min_{V(A)} \left\{ \frac{1}{8}(V(A)-0)^2 + \frac{1}{8}\times 0 + \cdots + \frac{1}{8}\times 0 \right\}$$

$$\Rightarrow \quad V(A) = 0$$

$$\min_{V(A)} \left\{ \frac{1}{8}(V(B)-0)^2 + \frac{1}{8}(V(B)-1)^2 + \cdots \right.$$
$$\left. \cdots + \frac{1}{8}(V(B)-1)^2 + \frac{1}{8}(V(B)-0) \right\}$$

$$\Rightarrow \quad V(B) = \frac{0+1+\cdots+1+0}{8} = \frac{6}{8}$$

- TD:

$\{S_t, A_t, R_{t+1}, S_{t+1}\}$.

$V(S_t)$ doesn't depend on past (Markov property)

S_t, A_t, R_{t+1}, S_{t+1}, A_{t+1}, R_{t+2}, S_{t+2}

A, 0, B, 0, L

B, 1, W

B, 1, W

B, 1, W

B, 1, W

B, 1, W

B, 1, W

B, 0, L

$\{S_t, A_t\} \rightarrow \{R_{t+1}, S_{t+1}\}$

$$p(s'|s,a) = \frac{N(s,a,s')}{N(s,a)} = \frac{1}{N(s,a)} \sum_{k=1}^{K=8} \sum_{t=1}^{T_k} I\{s_t^k, a_t^k, s_{t+1}^k = s,a,s'\}$$

$$p(r|s,a) = \frac{N(s,a,r)}{N(s,a)} = \frac{1}{N(s,a)} \sum_{k=1}^{K=8} \sum_{t=1}^{T_k} I\{s_t^k, a_t^k, r_{t+1}^k = s,a,r\}$$

$$R_s^a = \frac{1}{N(s,a)} \sum_{k=1}^{K=8} \sum_{t=1}^{T_k} I\{s_t^k, a_t^k, r_{t+1}^k = s,a,r\} r = \frac{1}{N(s,a)} \sum_{k=1}^{K=8} \sum_{t=1}^{T_k} I\{s_t^k, a_t^k = s,a\} r_{t+1}^k$$

$$\Rightarrow V(B) = 6/8 = 0.75.$$

$$V(A) = R_A + \gamma V(B)$$

$$= 0 + 1 \times V(B)$$

$$= 0.75.$$

Certainty Equivalence

- MC converges to solution with minimum mean-squared error
  - Best fit to the observed returns

$$\sum_{k=1}^{K}\sum_{t=1}^{T_k}\left(G_t^k - V(s_t^k)\right)^2 \times \frac{1}{8}$$

  - In the AB example, $V(A) = 0$ $V(B) = 6/8 = 0.75$
- TD(0) converges to solution of max likelihood Markov model
  - Solution to the MDP $\langle \mathcal{S}, \mathcal{A}, \hat{\mathcal{P}}, \hat{\mathcal{R}}, \gamma \rangle$ that best fits the data

$$\hat{\mathcal{P}}_{s,s'}^a = \frac{1}{N(s,a)}\sum_{k=1}^{K}\sum_{t=1}^{T_k}\mathbf{1}(s_t^k, a_t^k, s_{t+1}^k = s, a, s')$$

$$\hat{\mathcal{R}}_s^a = \frac{1}{N(s,a)}\sum_{k=1}^{K}\sum_{t=1}^{T_k}\mathbf{1}(s_t^k, a_t^k = s, a) r_t^k$$

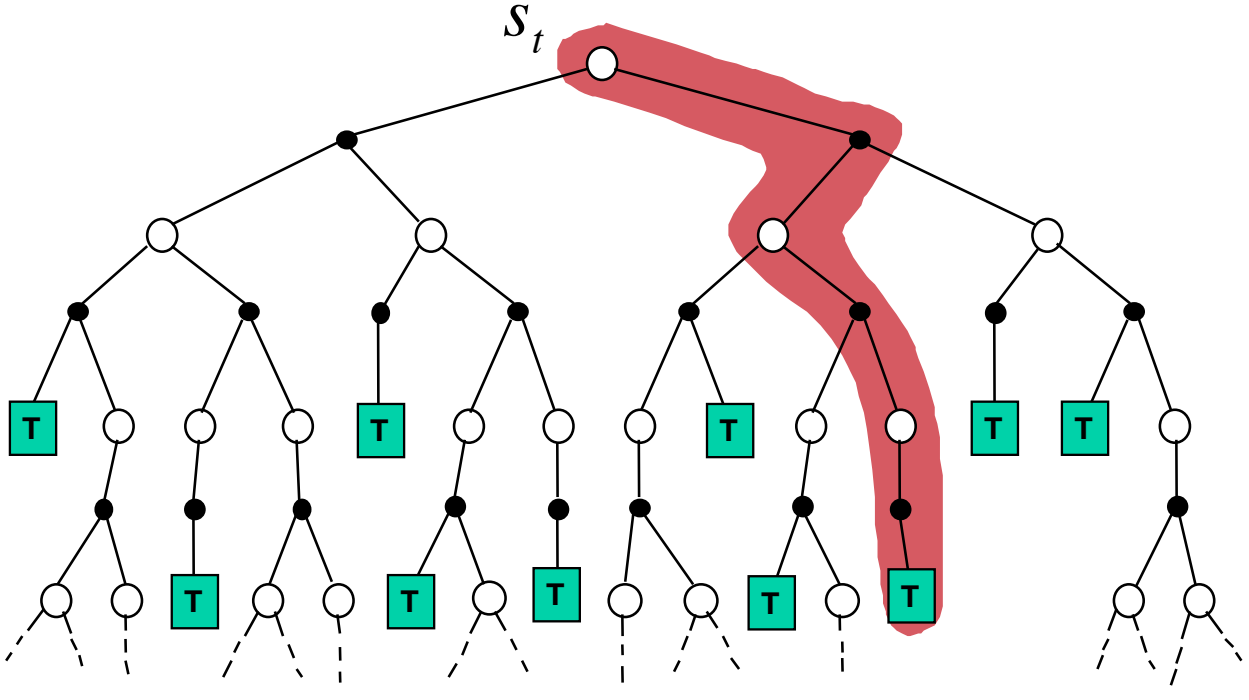
  - In the AB example, $V(A) = 0.75$, $V(B) = 6/8 = 0.75$

Advantages and Disadvantages of MC vs. TD (3)

- TD exploits Markov property
  - Usually more efficient in Markov environments
- MC does not exploit Markov property
  - Usually more effective in non-Markov environments

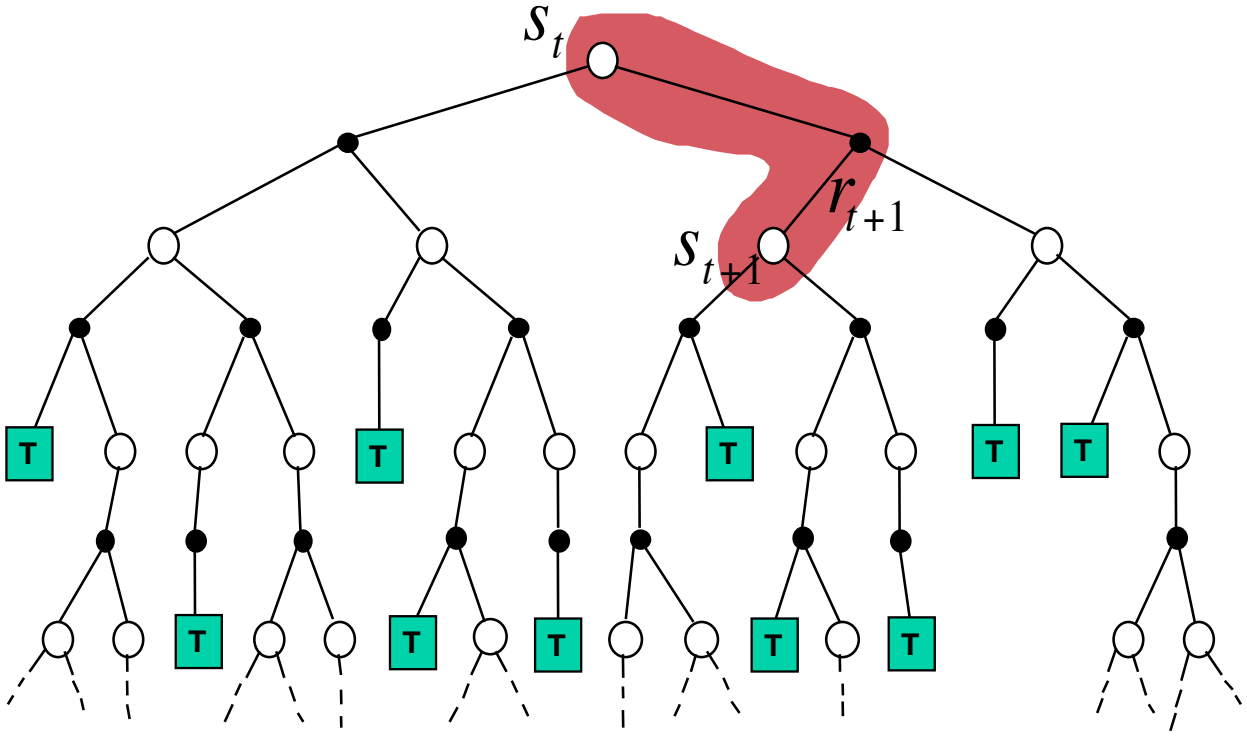
Monte-Carlo Backup

$$V(S_t) \leftarrow V(S_t) + \alpha\left(G_t - V(S_t)\right)$$



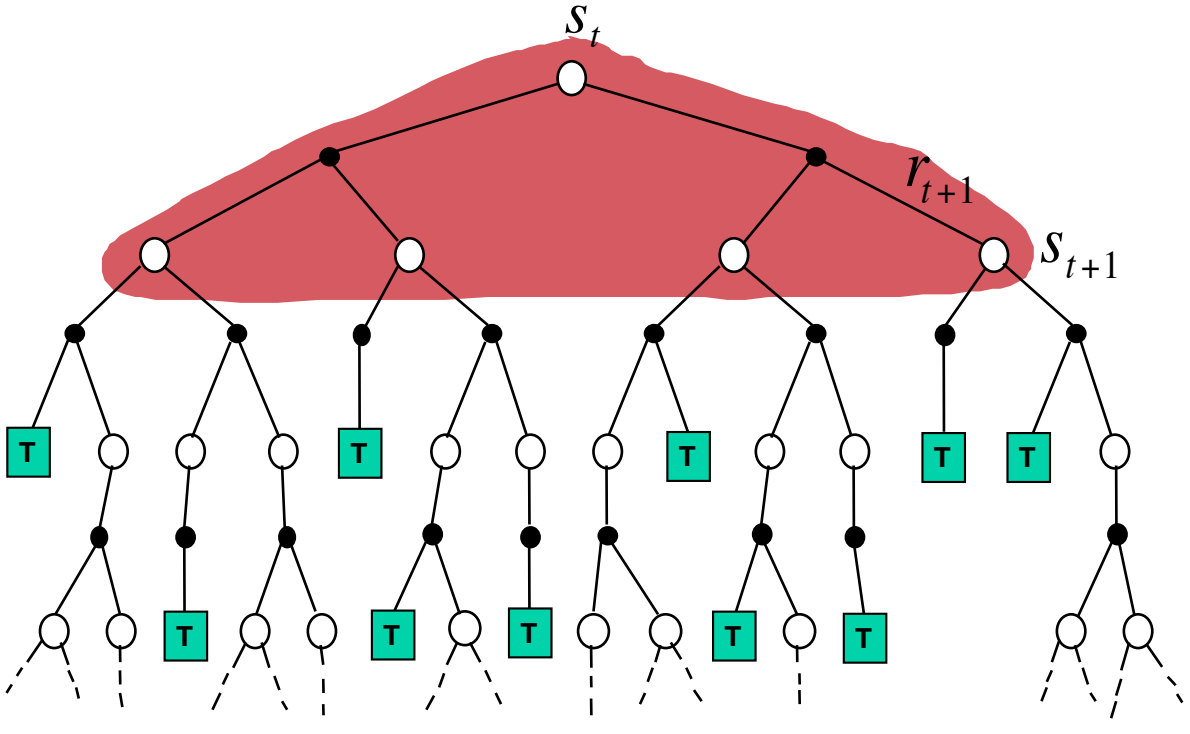
Temporal-Difference Backup

$$V(S_t) \leftarrow V(S_t) + \alpha\left(R_{t+1} + \gamma V(S_{t+1}) - V(S_t)\right)$$



Dynamic Programming Backup

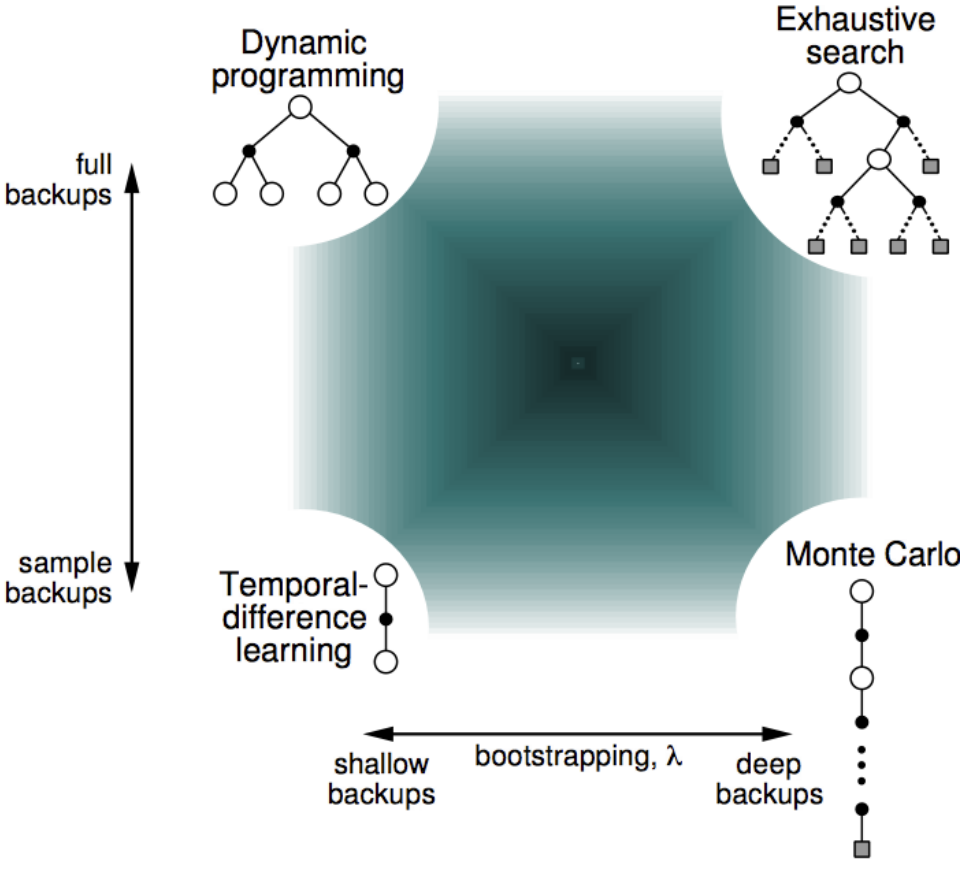
$$V(S_t) \leftarrow \mathbb{E}_\pi \left[R_{t+1} + \gamma V(S_{t+1}) \right]$$



Bootstrapping and Sampling

- Bootstrapping: update involves an estimate
  - MC does not bootstrap
  - DP bootstraps
  - TD bootstraps
- Sampling: update samples an expectation
  - MC samples
  - DP does not sample
  - TD samples

Unified View of Reinforcement Learning

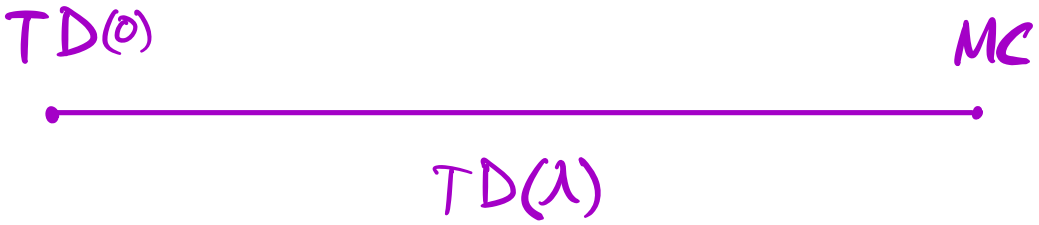


Example :

Shenzhen $\xrightarrow{\text{drive}}$ Beijing

- MC & TD:

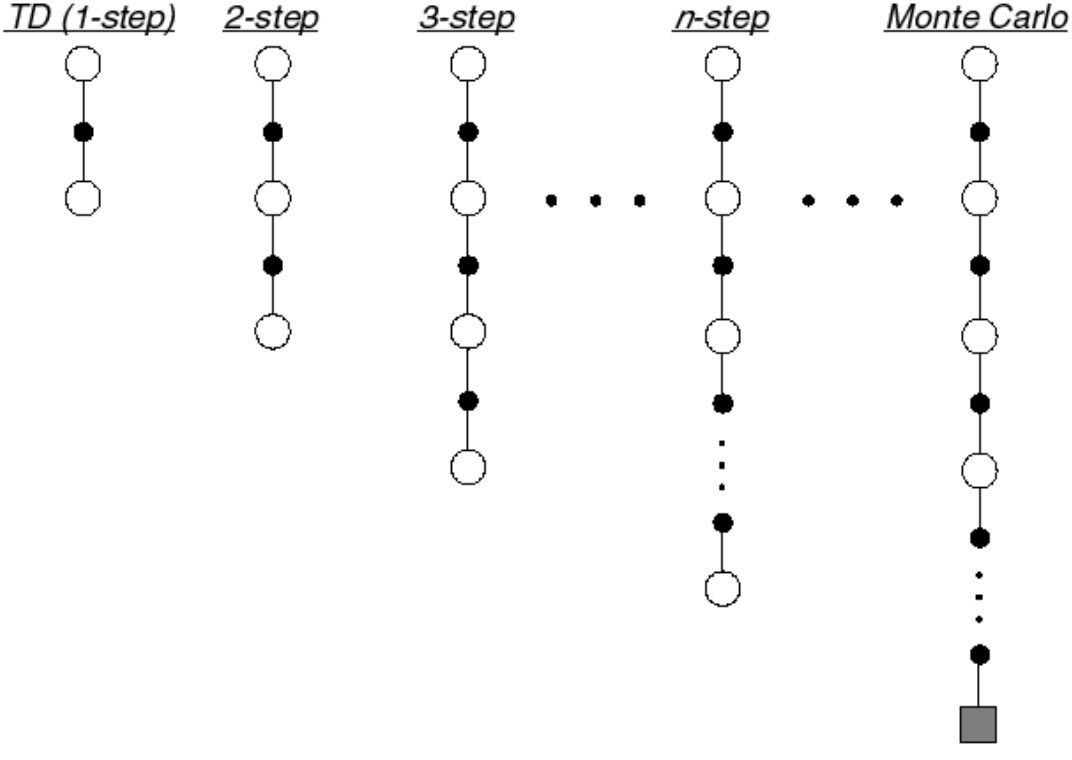
  two ends of the spetrum.



- Many algo in between.

n-Step Prediction

- Let TD target look n steps into the future



n-Step Return

- Consider the following n-step returns for $n = 1, 2, \infty$:

$$\begin{array}{lll} n = 1 & (TD) & G_t^{(1)} = R_{t+1} + \gamma V(S_{t+1}) \\ n = 2 & & G_t^{(2)} = R_{t+1} + \gamma R_{t+2} + \gamma^2 V(S_{t+2}) \\ \vdots & & \vdots \\ n = \infty & (MC) & G_t^{(\infty)} = R_{t+1} + \gamma R_{t+2} + ... + \gamma^{T-1} R_T \end{array}$$

- Define the n-step return

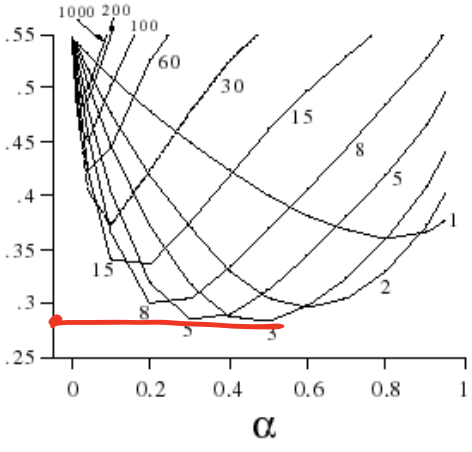
$$G_t^{(n)} = R_{t+1} + \gamma R_{t+2} + ... + \gamma^{n-1} R_{t+n} + \gamma^n V(S_{t+n})$$

- n-step temporal-difference learning

$$V(S_t) \leftarrow V(S_t) + \alpha \left(G_t^{(n)} - V(S_t) \right)$$

Large Random Walk Example

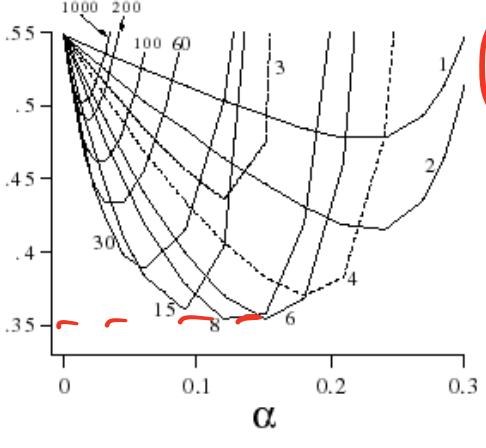
RMS error, averaged over first 10 episodes



ON-LINE n-STEP TD = Backward view

better

RMS error, averaged over first 10 episodes



OFF-LINE n-STEP TD = forward view

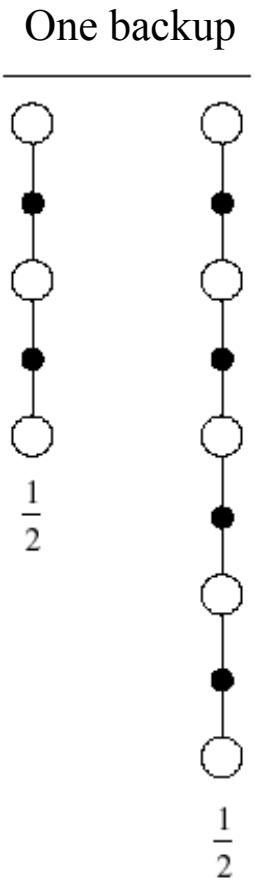
Averaging n-Step Returns

- We can average n-step returns over different n
- e.g. average the 2-step and 4-step returns

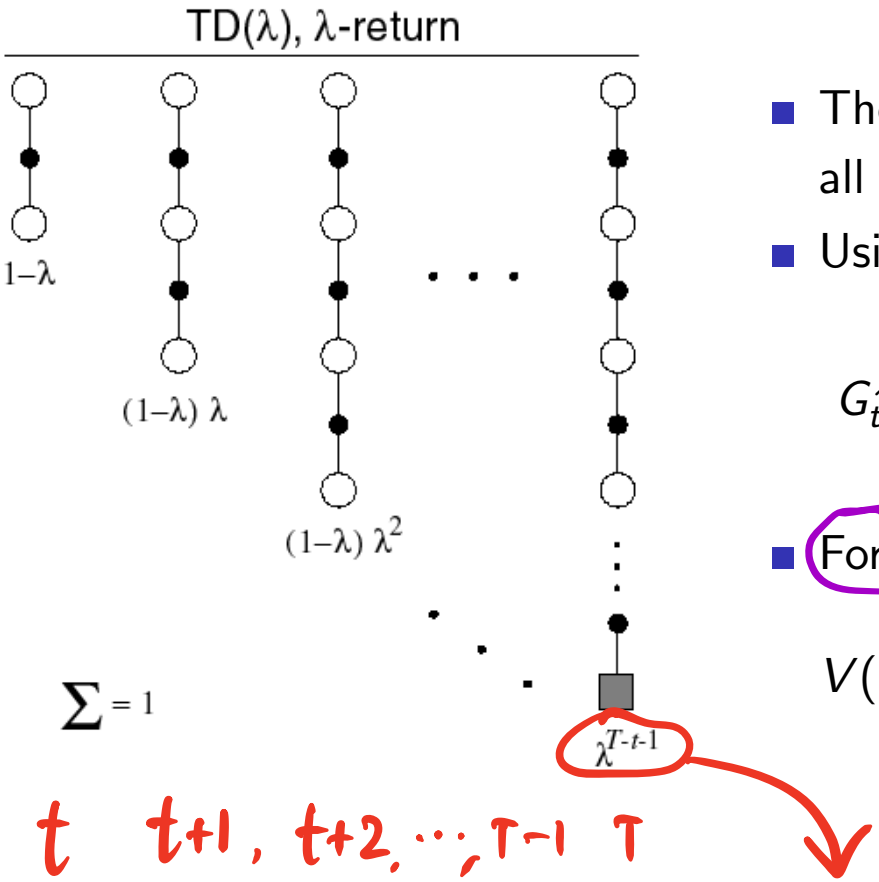
$$\frac{1}{2}G^{(2)} + \frac{1}{2}G^{(4)}$$

- Combines information from two different time-steps
- Can we efficiently combine information from all time-steps?

How to choose weight ?



λ-return



- The λ-*return* G_t^λ combines all n-step returns $G_t^{(n)}$
- Using weight $(1-\lambda)\lambda^{n-1}$

$$G_t^\lambda = (1-\lambda)\sum_{n=1}^{\infty} \lambda^{n-1} G_t^{(n)}$$

- Forward-view TD(λ)

$$V(S_t) \leftarrow V(S_t) + \alpha\left(G_t^\lambda - V(S_t)\right)$$

$$(1-\lambda)+(1-\lambda)\lambda+\cdots+(1-\lambda)\lambda^{T-t-2}+?=1$$

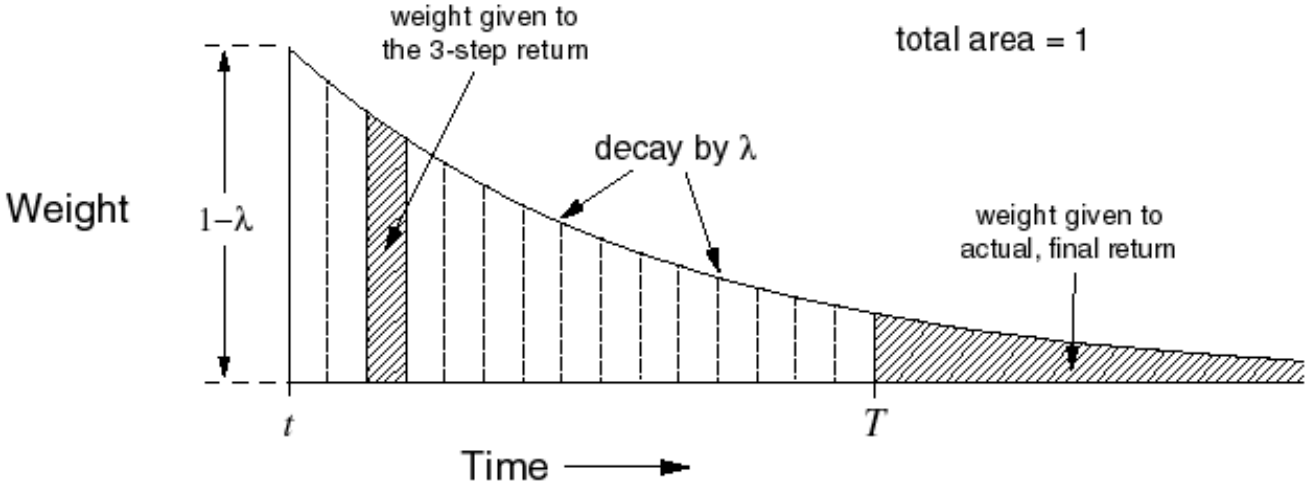
$$\Rightarrow$$

$$? = 1-(1-\lambda)\left[1+\lambda+\cdots+\lambda^{T-t-2}\right]$$

$$= 1-(1-\lambda)\frac{1-\lambda^{T-t-1}}{1-\lambda}$$

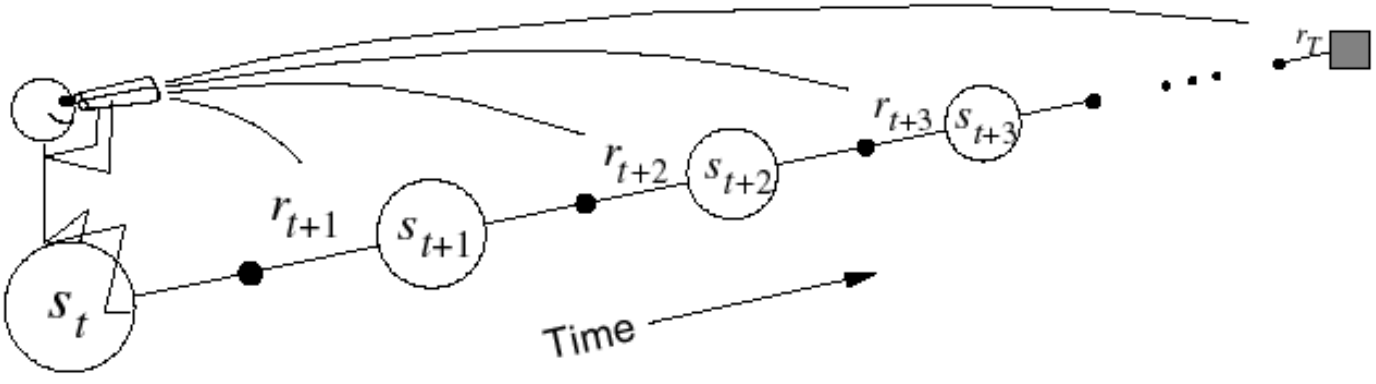
$$= \lambda^{T-t-1}.$$

TD(λ) Weighting Function



$$G_t^\lambda = (1-\lambda)\sum_{n=1}^{\infty} \lambda^{n-1} G_t^{(n)}$$

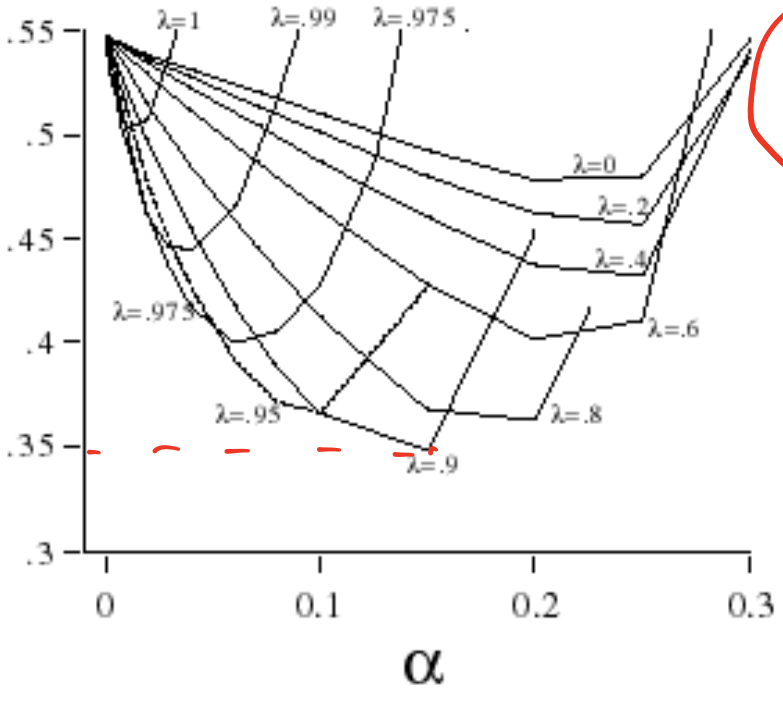
Forward-view TD(λ)



- Update value function towards the λ-return
- Forward-view looks into the future to compute G_t^λ
- Like MC, can only be computed from complete episodes

Forward-View TD(λ) on Large Random Walk

RMS error, averaged over first 10 episodes



OFF-LINE
λ-RETURN

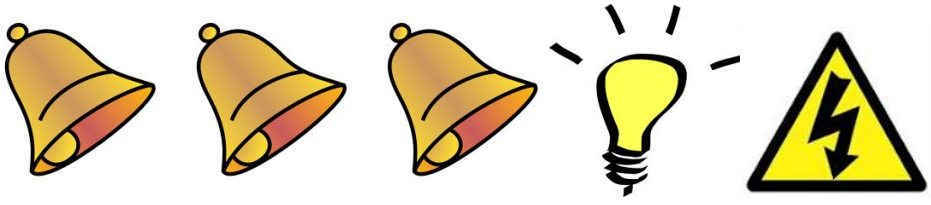
Forward-view

Backward View TD(λ)

- Forward view provides theory
- Backward view provides mechanism
- Update online, every step, from incomplete sequences

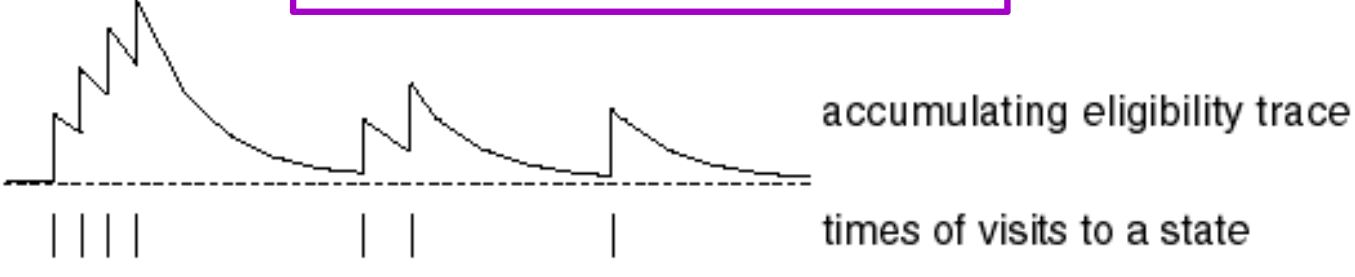


Eligibility Traces



- Credit assignment problem: did bell or light cause shock?
- Frequency heuristic: assign credit to most frequent states
- Recency heuristic: assign credit to most recent states
- *Eligibility traces* combine both heuristics

$$E_0(s) = 0$$
$$E_t(s) = \gamma\lambda E_{t-1}(s) + \mathbf{1}(S_t = s)$$

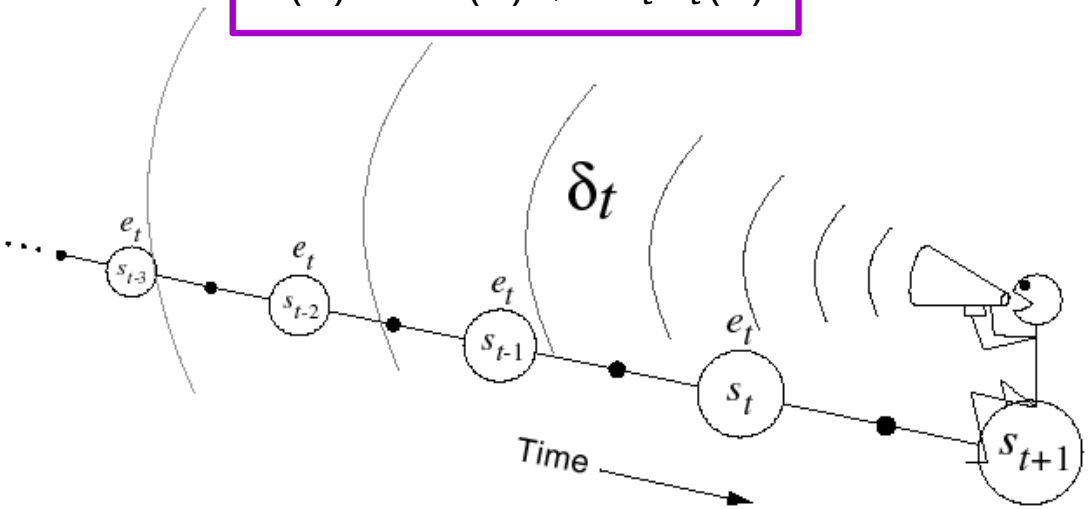


Backward View TD(λ)

- Keep an eligibility trace for every state s
- Update value $V(s)$ for every state s
- In proportion to TD-error δ_t and eligibility trace $E_t(s)$

$$\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$$

$$V(s) \leftarrow V(s) + \alpha \delta_t E_t(s)$$



TD(λ) and TD(0)

- When $\lambda = 0$, only current state is updated

$$E_t(s) = \mathbf{1}(S_t = s)$$
$$V(s) \leftarrow V(s) + \alpha \delta_t E_t(s)$$

- This is exactly equivalent to TD(0) update

$$V(S_t) \leftarrow V(S_t) + \alpha \delta_t$$

Recall : $V(S_t) \leftarrow V(S_t) + \alpha \left[R_{t+1} + \gamma V(S_{t+1}) - V(S_t) \right]$

TD(λ) and MC

- When $\lambda = 1$, credit is deferred until end of episode
- Consider episodic environments with offline updates
- Over the course of an episode, total update for TD(1) is the same as total update for MC

Theorem

The sum of offline updates is identical for forward-view and backward-view TD(λ)

$$\sum_{t=1}^{T} \alpha \delta_t E_t(s) = \sum_{t=1}^{T} \alpha \left(G_t^\lambda - V(S_t) \right) \mathbf{1}(S_t = s)$$

Forward-view

Backward-view.



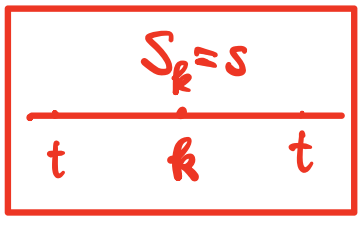
Exercise !

MC and TD(1)

- Consider an episode where s is visited once at time-step k,
- TD(1) eligibility trace discounts time since visit,

$$E_t(s) = \gamma E_{t-1}(s) + \mathbf{1}(S_t = s)$$
$$= \begin{cases} 0 & \text{if } t < k \\ \gamma^{t-k} & \text{if } t \geq k \end{cases}$$

$E_1(s) = \cdots = E_{k-1}(s) = 0$
$E_k(s) = 1$
$E_{k+1}(s) = \gamma E_k(s) = \gamma$
$E_{k+2}(s) = \gamma E_{k+1}(s) = \gamma^2$, ...



$\mathbb{I}(S_t = s) = \begin{cases} 1 & t = k \\ 0 & t \neq k \end{cases}$

- TD(1) updates accumulate error *online*

$$\sum_{t=1}^{T-1} \alpha \delta_t E_t(s) = \alpha \sum_{t=k}^{T-1} \gamma^{t-k} \delta_t = \alpha \left(G_k - V(S_k) \right)$$

- By end of episode it accumulates total error

$$\delta_k + \gamma \delta_{k+1} + \gamma^2 \delta_{k+2} + \ldots + \gamma^{T-1-k} \delta_{T-1}$$

Telescoping in TD(1)

When $\lambda = 1$, sum of TD errors telescopes into MC error,

$$
\begin{aligned}
&\delta_t + \gamma\delta_{t+1} + \gamma^2\delta_{t+2} + ... + \gamma^{T-1-t}\delta_{T-1} \\
&= R_{t+1} + \gamma V(S_{t+1}) - V(S_t) \\
&+ \gamma R_{t+2} + \gamma^2 V(S_{t+2}) - \gamma V(S_{t+1}) \\
&+ \gamma^2 R_{t+3} + \gamma^3 V(S_{t+3}) - \gamma^2 V(S_{t+2}) \\
&\quad \vdots \\
&+ \gamma^{T-1-t} R_T + \gamma^{T-t} V(S_T) - \gamma^{T-1-t} V(S_{T-1}) \\
&= R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} ... + \gamma^{T-1-t} R_T - V(S_t) \\
&= G_t - V(S_t)
\end{aligned}
$$

TD(λ) and TD(1)

- TD(1) is roughly equivalent to every-visit Monte-Carlo
- Error is accumulated online, step-by-step
- If value function is only updated offline at end of episode
- Then total update is exactly the same as MC

Telescoping in TD(λ)

For general λ, TD errors also telescope to λ-error, $G_t^\lambda - V(S_t)$

$$
\begin{aligned}
G_t^\lambda - V(S_t) &= -V(S_t) \quad + \quad (1-\lambda)\lambda^0 \left(R_{t+1} + \gamma V(S_{t+1})\right) \\
&\quad + \quad (1-\lambda)\lambda^1 \left(R_{t+1} + \gamma R_{t+2} + \gamma^2 V(S_{t+2})\right) \\
&\quad + \quad (1-\lambda)\lambda^2 \left(R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \gamma^3 V(S_{t+3})\right) \\
&\quad + \quad \ldots \\
&= -V(S_t) \quad + \quad (\gamma\lambda)^0 \left(R_{t+1} + \gamma V(S_{t+1}) - \gamma\lambda V(S_{t+1})\right) \\
&\quad + \quad (\gamma\lambda)^1 \left(R_{t+2} + \gamma V(S_{t+2}) - \gamma\lambda V(S_{t+2})\right) \\
&\quad + \quad (\gamma\lambda)^2 \left(R_{t+3} + \gamma V(S_{t+3}) - \gamma\lambda V(S_{t+3})\right) \\
&\quad + \quad \ldots \\
&= \quad (\gamma\lambda)^0 \left(R_{t+1} + \gamma V(S_{t+1}) - V(S_t)\right) \\
&\quad + \quad (\gamma\lambda)^1 \left(R_{t+2} + \gamma V(S_{t+2}) - V(S_{t+1})\right) \\
&\quad + \quad (\gamma\lambda)^2 \left(R_{t+3} + \gamma V(S_{t+3}) - V(S_{t+2})\right) \\
&\quad + \quad \ldots \\
&= \delta_t + \gamma\lambda\delta_{t+1} + (\gamma\lambda)^2\delta_{t+2} + \ldots
\end{aligned}
$$

Forwards and Backwards TD(λ)

- Consider an episode where s is visited once at time-step k,
- TD(λ) eligibility trace discounts time since visit,

$$\begin{aligned} E_t(s) &= \gamma\lambda E_{t-1}(s) + \mathbf{1}(S_t = s) \\ &= \begin{cases} 0 & \text{if } t < k \\ (\gamma\lambda)^{t-k} & \text{if } t \geq k \end{cases} \end{aligned}$$

- Backward TD(λ) updates accumulate error *online*

$$\sum_{t=1}^{T} \alpha\delta_t E_t(s) = \alpha \sum_{t=k}^{T} (\gamma\lambda)^{t-k}\delta_t = \alpha\left(G_k^\lambda - V(S_k)\right)$$

- By end of episode it accumulates total error for λ-return
- For multiple visits to s, $E_t(s)$ accumulates many errors

Offline Equivalence of Forward and Backward TD

Offline updates

- Updates are accumulated within episode
- but applied in batch at the end of episode

Onine Equivalence of Forward and Backward TD

Online updates

- TD(λ) updates are applied online at each step within episode
- Forward and backward-view TD(λ) are slightly different
- NEW: Exact online TD(λ) achieves perfect equivalence
- By using a slightly different form of eligibility trace
- Sutton and von Seijen, ICML 2014

Summary of Forward and Backward TD(λ)

Offline updates	$\lambda = 0$	$\lambda \in (0, 1)$	$\lambda = 1$
Backward view	TD(0)	TD(λ)	TD(1)
	$\parallel$	$\parallel$	$\parallel$
Forward view	TD(0)	Forward TD(λ)	MC
Online updates	$\lambda = 0$	$\lambda \in (0, 1)$	$\lambda = 1$
Backward view	TD(0)	TD(λ)	TD(1)
	$\parallel$	$\not\parallel$	$\not\parallel$
Forward view	TD(0)	Forward TD(λ)	MC
	$\parallel$	$\parallel$	$\parallel$
Exact Online	TD(0)	Exact Online TD(λ)	Exact Online TD(1)

$=$ here indicates equivalence in total update at end of episode.

Suffices to show:

$$\sum \left[R_{t+1} + \gamma V(S_{t+1}) - V(S_t)\right] \cdot \left[\gamma E_{t-1}(s) + I(S_t = s)\right]$$

$$= \sum \left[G_t^T - V(S_t)\right] I(S_t = s)$$